

INSERTION SORT

SIMATS Online Programming Lab

DS PROGRAMMING

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PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,i,j,key;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=1;i<n;i++)
9     {
10        key=a[i];
11        j=i-1;
12        while(j>=0 && a[j]>key)
13        {
14            a[j+1]=a[j];
15            j--;
16        }
17        a[j+1]=key;
18    }
19    for(i=0;i<n;i++)
20        printf(" %d",a[i]);
21    return 0;
```

INPUT

5
40 10 20 30 50

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```
17     a[j+1]=key;
18 }
19 for(i=0;i<n;i++)
20     printf(" %d",a[i]);
21 return 0;
22 }
23
24
```

OUTPUT

10 20 30 40 50

